

Supplemental Material for: Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

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I. ARCHIVED LEGACY-PBE TETRAMER BASELINE

Table I reproduces the archived rigid-strain grid (PBE without D3) retained for reproducibility; it is *not* used for quantitative claims in the main text.

II. CALCULATION INVENTORY (REPRODUCIBILITY)

Tables II and III list the DFT groups underlying the main-text claims and superseded exploratory grids retained only for reproducibility. They are *not* scientific results tables.

III. FULL TETRAMER π -DOS AT $\epsilon=0$

IV. PERIODIC FRONTIER-ORBITAL LOCALIZATION

Cage-resolved VBM/CBM isosurfaces on periodic $n \times C_{60}$ cells ($n = 2-8$) at $\epsilon=0$ complement the tetramer π -DOS above and the main-text electronic taxonomy (donor/acceptor vs. size dilution). Carbon cages are shown as uniform gray MSMS molecular surfaces (probe 1.3 Å); heteroatoms use element colors.

V. PERIODIC SYNERGY ROWS ($n \geq 6$)

VI. MARCUS REORGANIZATION ENERGIES

Polaron transport diagnostics are reported here for completeness; the main-text analysis centers on S .

VII. HOPPING INTEGRAL J

On the tetramer dimer grid, B substitution at $\epsilon=0$ delocalizes the charged-state orbital (IPR 75.0 $\rightarrow$ 24.6) while raising J from 26.6 to 46.4 meV; coupled B @ +3% gives $J = 37.2$ meV and IPR 23.1. The criterion $J > \lambda/2$ for polaron hopping dominance [1] is *not* met at PBE+D3 ($J < \lambda/2$), so a polaron transition is not confirmed on this grid.

[1] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa,

electron localization and mobility in monolayer fullerene networks, Nano Lett. **24**, 7891 (2024).

TABLE I. Archived legacy-PBE (no D3) tetramer rigid-strain baseline retained for reproducibility only; not used for main-text quantitative claims. $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at $\epsilon=0$ (no chemical potentials; relative ranking within this protocol only).

System	n_{dop}	$E(\epsilon=0)$ (E_{h})	$E(\epsilon=+5\%)$ (E_{h})	α (meV/%)	E_{sub} (eV/dopant)
Pristine	—	−1363.5512	−1363.5509	−1.0	—
B-doped	13	−1328.4900	−1328.4090	+56.9	+73.4
N-doped	12	−1414.3812	−1414.4053	−303.2	−115.3
P-doped	12	−1370.4965	−1370.4888	−19.1	−15.8

TABLE II. DFT calculation groups supporting the main-text claims (reproducibility inventory; Supplemental Material only). Superseded exploratory grids appear in Table III.

Calculation group	System	Count	Protocol
Archived baseline (legacy PBE)	$4 \times \text{C}_{60}$	24	Rigid strain; not used for main-text claims
π -DOS and gaps	$4 \times \text{C}_{60}$	12	PBE+D3
IPR (polaron dimer)	Dimer	4	Pristine/B; $\epsilon=0, +3\%$
Hopping J	Dimer	4	PBE+D3
Charged polarons ^a	$4 \times \text{C}_{60}$	20	UKS; geometry opt. + vertical SP
Periodic size scaling	$n \times \text{C}_{60}$	41	PBE+D3; 400/350 Ry
Cutoff benchmark	$6 \times \text{C}_{60}$ N	1	+3%; 400 Ry single-point
Ionic relaxation	$4 \times \text{C}_{60}$	4	Fixed-cell GEO; P corners
Ionic relaxation	$1 \times \text{C}_{60}$ P	4	Fixed-cell GEO; periodic
Alternate placement	$4 \times \text{C}_{60}$	24	Rigid PBE+D3
Periodic placement	$4 \times \text{C}_{60}$	16	$n=4$; two independent maps
Matched-functional rigid	$4 \times \text{C}_{60}$	4	PBE+D3 at relaxed coordinates
Reference placement	$4 \times \text{C}_{60}$	24	Rigid PBE+D3
Hirshfeld population	$1 \times \text{C}_{60}$ B/N/P	18	Spin-polarized KS

^aSix relaxed tetramer geometries provide charged-state initial structures.

TABLE III. Exploratory DFT grids superseded by the production protocol (reproducibility archive only).

Task group	System	N_{conv}	Disposition
Early strain probe	$2 \times \text{C}_{60}$	5	Superseded
Single-molecule doping	C_{60}	11	Superseded
Early electronics	$2 \times \text{C}_{60}$	24	Superseded
Exploratory design grid	$4 \times \text{C}_{60}$	20	Archive only

TABLE IV. Supplemental periodic synergy rows at +3% ($n \geq 6$; 350 Ry production cutoff; completeness only).

n	Dopant	ϵ (%)	$\mathcal{S}$ (meV/atom)
6	B	+3	−7.76
6	N	+3	+2.38
6	P	+3	−31.89
8	B	+3	−11.26
8	N	+3	−2.30
8	P	+3	−29.94

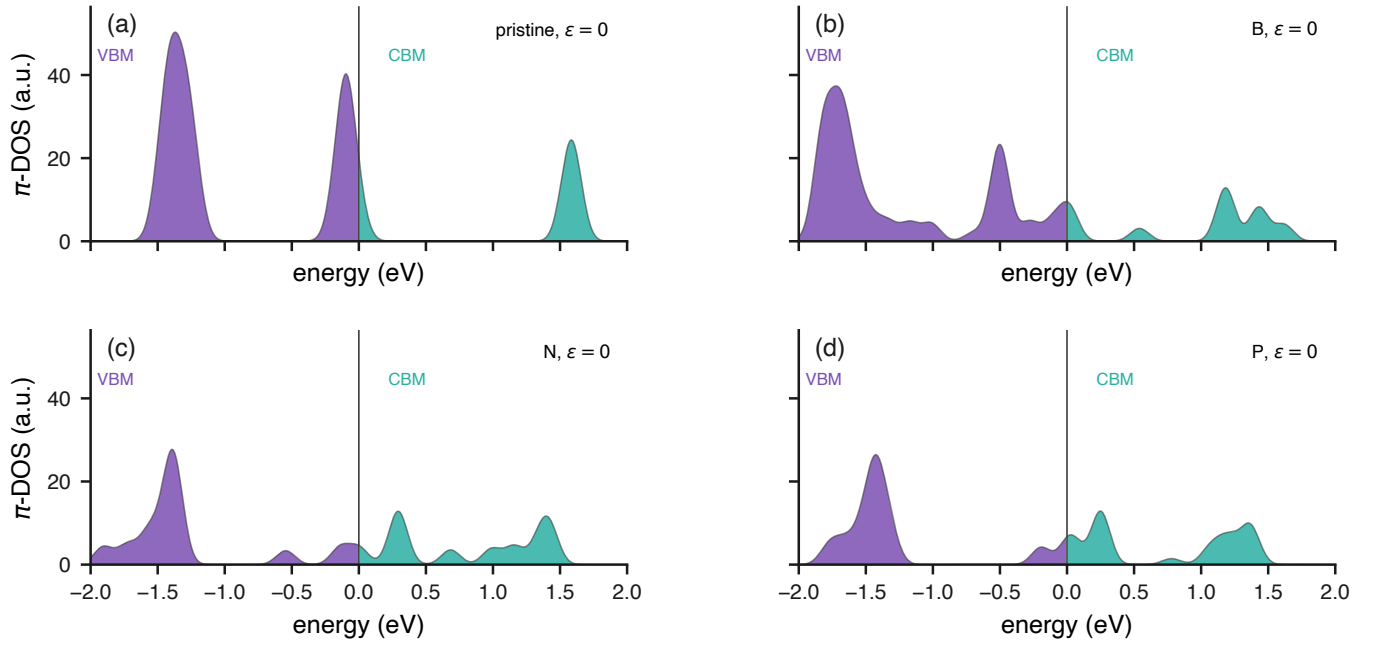
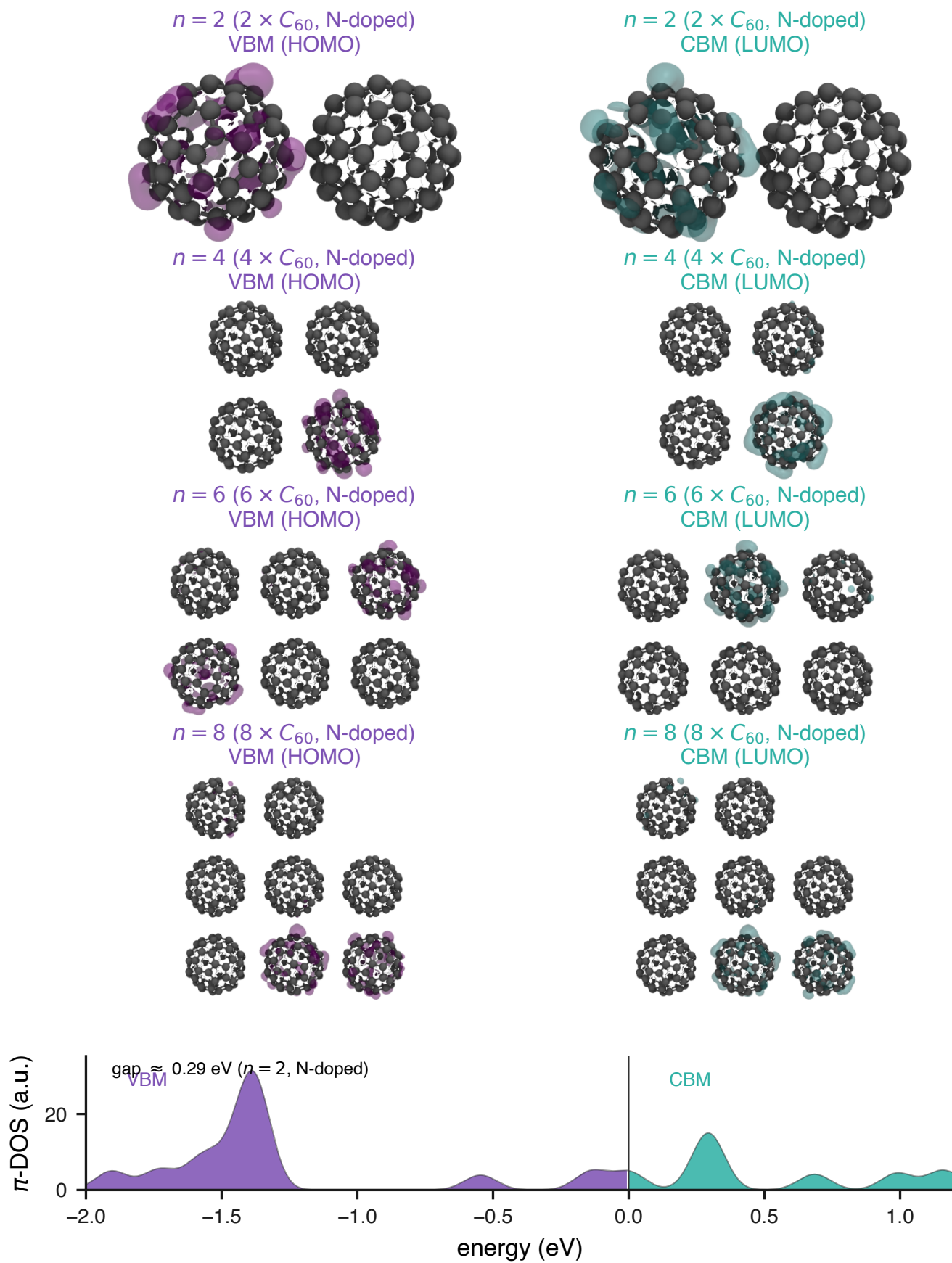


FIG. 1. Full π -DOS panels for pristine and B/N/P-doped tetramers at $\epsilon=0$ (PBE+D3), complementary to the main-text electronic-structure figure.

qHP C_{60} graphullerene ($\varepsilon = 0\%$, N-doped)

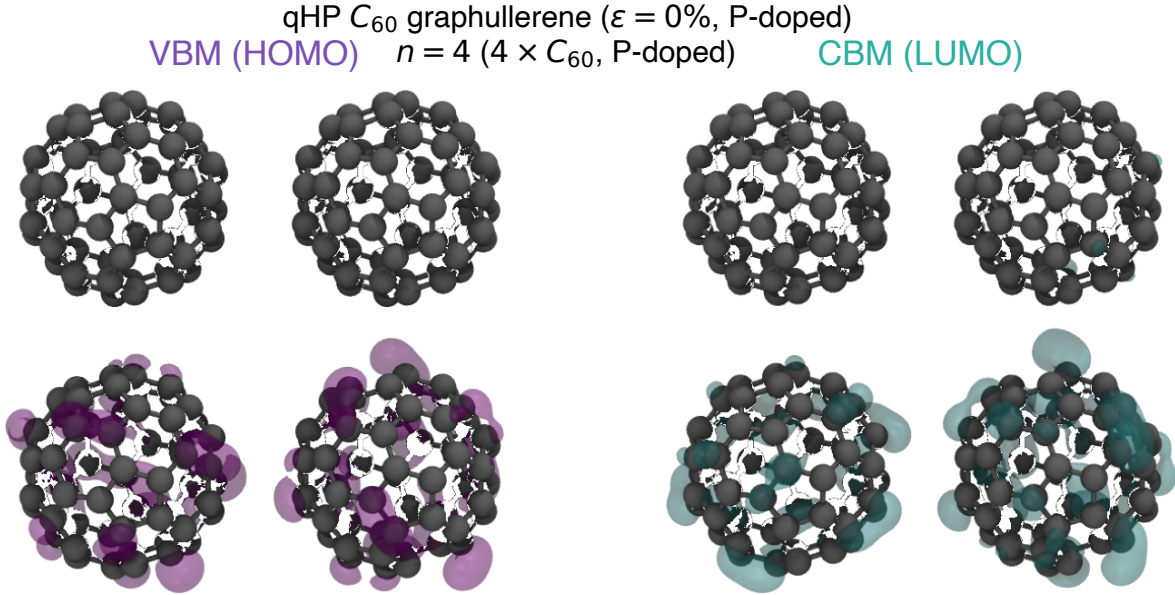


FIG. 3. P-doped $n=4$ supercell at $\epsilon=0$: VBM (left) and CBM (right) on gray MSMS cages. Both frontier states localize on a subset of cages, providing a spatial counterpart to the large periodic $|S|$ of size-mismatched P reported in the main text.

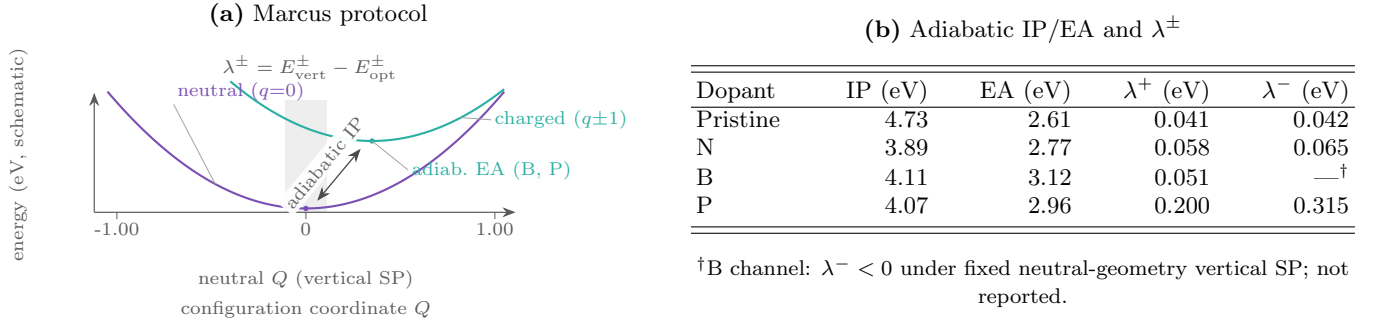


FIG. 4. Marcus reorganization energies from charged-state geometry optimizations and vertical single-points at the neutral geometry. Panel (a) schematically defines the vertical-adiabatic decomposition following Ref. 1; panel (b) tabulates adiabatic IP/EA and $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$ for pristine and B/N/P tetramers. The electron-channel λ^- for B is unphysical under the fixed neutral-geometry vertical protocol and is excluded (Limitations).

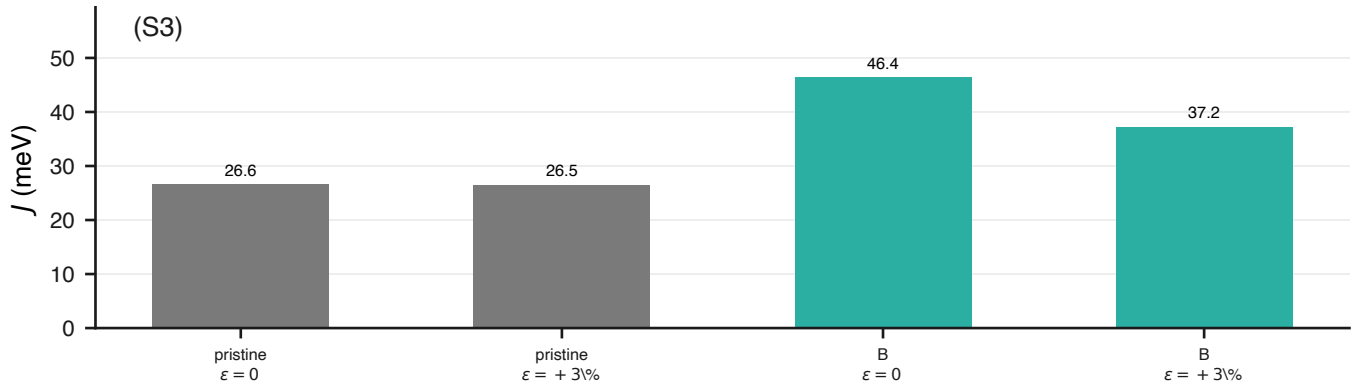


FIG. 5. Nearest-neighbor electronic coupling J on the tetramer dimer grid (pristine and B-doped cells at $\epsilon=0$ and $+3\%$; PBE+D3). Companion inverse-participation-ratio values for the same four corners are tabulated in the open data release (Methods).